

Marcus Lodge

Plate Stress Analysis

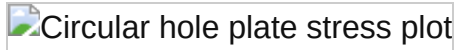


Plate 1: Circular Hole Stress Plot

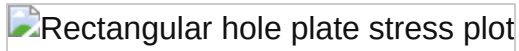


Plate 2: Rectangular Hole Stress Plot

The highest stress in the plate designs occurred in small areas within the interior edges of the holes in the plates. The plate with the circular hole, Plate 1, had the highest stress at the edge of the hole near the horizontal center of the plate. The two high-stress points were vertically symmetrical across the perpendicular axis through the center of the hole.

The plate with the rectangular hole, Plate 2, experienced moderate stress at four points within the interior of the hole. These four points were located adjacent to the corners of the rectangular hole on the faces parallel with the long edge of the plate. This stress also exhibited symmetrical patterns both horizontally and vertically across axes intersecting the middle of the hole.

It is important to note that the external force had a nearly identical effect on both plates, with one small deviation. Based on the plots, Plate 1 had an extremely high point of stress at the center edge of the hole. Alternatively, Plate 2 showed only moderate stress points. The stress within Plate 1 was distributed between two points, while Plate 2 had four points of higher stress. The greater quantity of stress points resulted in lower stress concentration at each individual point, while the overall stress effect remained similar between the two designs.